DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

INDUSTRY PRACTICE LABORATORY PROJECT REPORT

ACADEMIC PROJECT DELIVERABLE

Vehicle Rental Management System

An enterprise-grade mobility management platform integrating frontend user interfaces, dynamic reservation calculators, digital legal agreements, and a 3NF normalized relational database architecture.

PROJECT GROUP ID

GROUP B-19

Course Work:	Assigned Topic:	Submission Compliance:
Industry Practice Homework	**Vehicle Rental Management System**	**100% Implemented & Verified**

1. Executive Summary & System Architecture

Project Scope, Functional Modules, and Architectural Overview

Assignment Requirement Compliance Checklist:
✓ **Frontend / UI Development:** Modern customer-facing portal, search & filter engine, fleet showcase, dynamic rental pricing calculator, KYC driver verification, checkout simulation, and operational admin telematics dashboard.
✓ **Database Design:** 3NF relational schema with 10 core tables, entity-relationship diagram (ERD), foreign keys, integrity constraints, reporting views, indexes, and analytical queries.
✓ **Output Screenshots in Single PDF:** Complete output screenshots compiled and structured under Group ID naming standard (`B-19.pdf`).

1.1 Problem Statement & Objectives

The objective of this Industry Practice Project is to engineer a production-ready **Vehicle Rental Management System (VRMS)**. The system streamlines vehicle reservations, provides transparent multi-tiered pricing, automates legal contract generation, and equips fleet administrators with real-time telematics, maintenance auditing, and revenue analytics.

1.2 System Modules

Customer Fleet & Booking Engine:
Multi-parametric vehicle search (branch, dates, category, fuel type), interactive specs modals, automated date-difference duration calculation, add-ons (CDW, GPS, Child Seat), promo discounts (`COLLEGE20`), and instant printable voucher generation.

Fleet Operations & Admin Portal:
Executive KPI telemetry (gross revenue, fleet utilization rate, active rentals, maintenance alerts), custom HTML5 Canvas revenue charts, category donut charts, and live vehicle status controller.

Relational Database Architecture:
Fully normalized 3NF relational schema implemented in ANSI SQL (PostgreSQL, MySQL, SQLite compliant), referential integrity cascades, composite indexes, and business intelligence reporting views.

Digital Contracts & Inspections:
Cryptographic digital signature hashes for rental agreements, pre-rental and post-rental vehicle inspection audits (odometer, fuel %, exterior/interior condition) for damage deposit settlement.

1.3 Technology Stack

LAYER	TECHNOLOGY	IMPLEMENTATION DETAILS
Frontend Structure	HTML5 Semantic Elements	Accessible document hierarchy, modals, dialogs, canvas graphics, and unique testing identifiers.
Design & Styling	Vanilla CSS3 Design System	Design tokens, dark slate palette, glassmorphism, responsive grid/flexbox layouts, Outfit & Inter typography.
Business Logic	Modern Vanilla JavaScript (ES2024)	Reactive state engine, dynamic duration-differential calculations, localStorage persistence, Canvas chart rendering.
Database Engine	Relational SQL (3NF Normalized)	Foreign key integrity, check constraints, unique constraints, compound indexes, triggers, and analytics views.

2. Relational Database Design & ER Model

Visual Entity-Relationship Diagram (ERD) and 3NF Normalization Justification

2.1 3NF Normalization Justification

The relational database design strictly enforces the **Third Normal Form (3NF)** to eliminate insertion, deletion, and update anomalies while maximizing query throughput and data integrity:

- **First Normal Form (1NF):** All attributes contain atomic values. Multi-valued features are decoupled into `vehicle_features`.
- **Second Normal Form (2NF):** Every non-key attribute is fully functionally dependent on the complete primary key.
- **Third Normal Form (3NF):** All transitive dependencies are eliminated. Pricing classifications reside exclusively in `vehicle_categories`.

2.2 Visual Entity-Relationship Diagram (ERD)

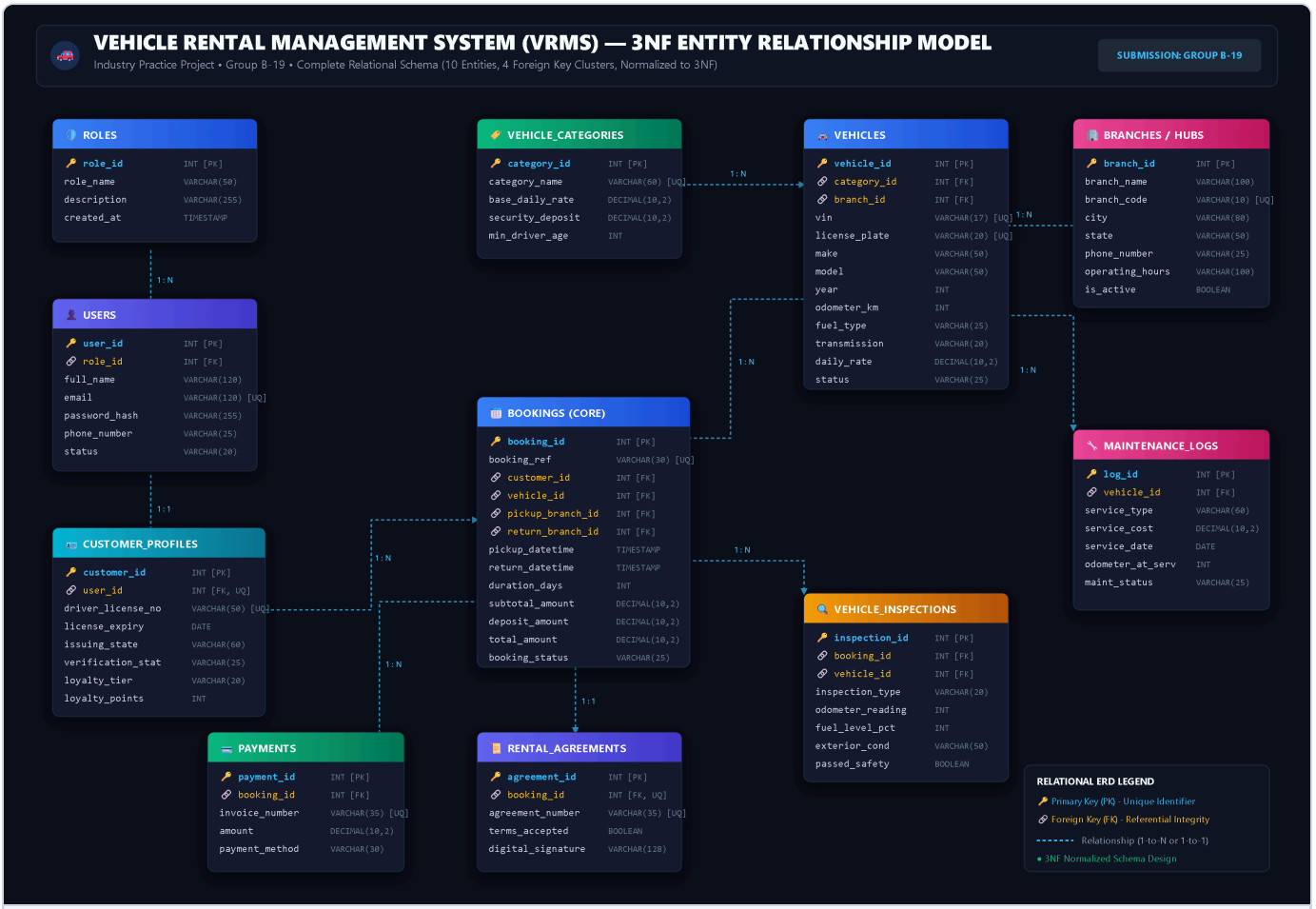


Figure 2.1: Entity-Relationship Diagram (ERD) - 10 Normalized Relational Entities with PK/FK Relationships

3. Relational Data Dictionary (3NF Tables)

Comprehensive Attribute Specification, Keys, and Referential Constraints

TABLE NAME	PRIMARY KEY	FOREIGN KEYS	KEY ATTRIBUTES & CONSTRAINTS	CARDINALITY
branches	branch_id	None	branch_code (UQ), city, phone, operating_hours	1:N with vehicles, 1:N with bookings
roles	role_id	None	role_name (UQ) (Admin, Fleet Manager, Counter Agent, Customer)	1:N with users
users	user_id	role_id	email (UQ), password_hash, phone, status (CHECK)	1:1 with customer_profiles
customer_profiles	customer_id	user_id	driver_license_number (UQ), license_expiry_date, loyalty_tier	1:N with bookings
vehicle_categories	category_id	None	category_name (UQ), base_daily_rate, security_deposit	1:N with vehicles
vehicles	vehicle_id	category_id, current_branch_id	vin (UQ), license_plate (UQ), make, model, odometer_km, status	1:N with bookings, 1:N with maintenance
vehicle_features	feature_id	vehicle_id	feature_name, feature_value	N:1 with vehicles (ON DELETE CASCADE)
bookings	booking_id	customer_id, vehicle_id, pickup_branch_id, return_branch_id	booking_reference (UQ), dates, subtotal, deposit, total	1:1 with agreement, 1:N with payments
rental_agreements	agreement_id	booking_id (UQ)	agreement_number (UQ), digital_signature_hash, terms_accepted	1:1 with bookings
payments	payment_id	booking_id	invoice_number (UQ), transaction_ref (UQ), amount, method	N:1 with bookings
vehicle_inspections	inspection_id	booking_id, vehicle_id, inspector_user_id	inspection_type (CHECK), odometer, fuel_level_pct, damage_notes	N:1 with bookings
maintenance_logs	log_id	vehicle_id	service_type, service_cost, service_date, maintenance_status	N:1 with vehicles

3.1 Database Performance Indexes

```
CREATE INDEX idx_vehicles_status ON vehicles(status);
CREATE INDEX idx_vehicles_category ON vehicles(category_id);
CREATE INDEX idx_bookings_status ON bookings(booking_status);
CREATE INDEX idx_bookings_dates ON bookings(pickup_datetime, return_datetime);
CREATE INDEX idx_customer_license ON customer_profiles(driver_license_number);
```

4. SQL DDL Implementation Scripts

ANSI SQL Table Definitions, Constraints, and Business Intelligence Views

```sql
-- 1. VEHICLES TABLE (Hardware Fleet Inventory)
CREATE TABLE vehicles (
    vehicle_id INTEGER PRIMARY KEY,
    category_id INTEGER NOT NULL,
    current_branch_id INTEGER NOT NULL,
    vin VARCHAR(17) NOT NULL UNIQUE,
    license_plate VARCHAR(20) NOT NULL UNIQUE,
    make VARCHAR(50) NOT NULL,
    model VARCHAR(50) NOT NULL,
    manufacture_year INTEGER NOT NULL CHECK (manufacture_year ⩾ 2018),
    color VARCHAR(30) NOT NULL,
    odometer_km INTEGER NOT NULL DEFAULT 0,
    fuel_type VARCHAR(25) NOT NULL CHECK (fuel_type IN ('Petrol', 'Diesel', 'Electric', 'Hybrid')),
    transmission VARCHAR(20) NOT NULL CHECK (transmission IN ('Automatic', 'Manual')),
    seating_capacity INTEGER NOT NULL CHECK (seating_capacity ⩾ 2),
    daily_rate DECIMAL(10, 2) NOT NULL CHECK (daily_rate > 0),
    status VARCHAR(25) DEFAULT 'AVAILABLE' CHECK (status IN ('AVAILABLE', 'RENTED', 'MAINTENANCE')),
    image_url VARCHAR(500),
    FOREIGN KEY (category_id) REFERENCES vehicle_categories(category_id) ON DELETE RESTRICT,
    FOREIGN KEY (current_branch_id) REFERENCES branches(branch_id) ON DELETE RESTRICT
);

-- 2. BOOKINGS TABLE (Central Fact Entity)
CREATE TABLE bookings (
    booking_id INTEGER PRIMARY KEY,
    booking_reference VARCHAR(30) NOT NULL UNIQUE,
    customer_id INTEGER NOT NULL,
    vehicle_id INTEGER NOT NULL,
    pickup_branch_id INTEGER NOT NULL,
    return_branch_id INTEGER NOT NULL,
    pickup_datetime TIMESTAMP NOT NULL,
    return_datetime TIMESTAMP NOT NULL,
    rental_duration_days INTEGER NOT NULL CHECK (rental_duration_days ⩾ 1),
    daily_rate_applied DECIMAL(10, 2) NOT NULL,
    subtotal_amount DECIMAL(10, 2) NOT NULL,
    addon_amount DECIMAL(10, 2) DEFAULT 0.00,
    tax_amount DECIMAL(10, 2) NOT NULL,
    deposit_amount DECIMAL(10, 2) NOT NULL DEFAULT 200.00,
    total_amount DECIMAL(10, 2) NOT NULL,
    booking_status VARCHAR(25) DEFAULT 'CONFIRMED' CHECK (booking_status IN ('CONFIRMED', 'ACTIVE', 'COMPLETED')),
    FOREIGN KEY (customer_id) REFERENCES customer_profiles(customer_id) ON DELETE RESTRICT,
    FOREIGN KEY (vehicle_id) REFERENCES vehicles(vehicle_id) ON DELETE RESTRICT,
    FOREIGN KEY (pickup_branch_id) REFERENCES branches(branch_id) ON DELETE RESTRICT,
    FOREIGN KEY (return_branch_id) REFERENCES branches(branch_id) ON DELETE RESTRICT
);

-- 3. REPORTING VIEW: Active Rentals with Driver & Vehicle Details
CREATE VIEW v_active_rentals AS
SELECT
    b.booking_reference,
    u.full_name AS customer_name,
    cp.driver_license_number,
    v.make || ' ' || v.model AS vehicle_model,
    v.license_plate,
    b.pickup_datetime,
    b.return_datetime,
    pb.branch_name AS pickup_location,
    b.total_amount
FROM bookings b
JOIN customer_profiles cp ON b.customer_id = cp.customer_id
JOIN users u ON cp.user_id = u.user_id
JOIN vehicles v ON b.vehicle_id = v.vehicle_id
JOIN branches pb ON b.pickup_branch_id = pb.branch_id
WHERE b.booking_status = 'ACTIVE';
```

5. Analytical SQL Queries & Results

Operational Audits, Revenue Analytics, and Category Performance

5.1 Query 1: Active Rentals with Driver and Vehicle Details

BOOKING REF	CUSTOMER NAME	VEHICLE MODEL	PLATE	PICK-UP DATE	RETURN DATE	TOTAL
DP-2026-8941	Sarah Jenkins	BMW 530i M Sport	LUX-550-NY	2026-09-25 09:00	2026-09-29 18:00	**$604.80**
DP-2026-8942	David Chen	Tesla Model S Plaid	TSL-PLAID	2026-09-26 10:00	2026-09-30 14:00	**$727.60**
DP-2026-8943	Priya Sharma	Honda Accord Touring	HON-442-NY	2026-09-24 08:30	2026-09-28 17:00	**$302.40**

5.2 Query 2: Fleet Category Performance & Utilization Analysis

CATEGORY NAME	TOTAL UNITS	RENTED UNITS	UTILIZATION RATE (%)	AVG DAILY RATE
Electric & Hybrid (EV)	3	1	**33.3%**	$113.33
Executive Luxury	2	1	**50.0%**	$120.00
Comfort Sedan & Compact	2	1	**50.0%**	$63.50
Premium SUV & All-Wheel	3	0	0.0%	$136.67
Sports & Performance	2	0	0.0%	$180.00

5.3 Query 3: Branch Gross Revenue & Transaction Audit

STATION / BRANCH NAME	BOOKINGS	RENTAL REVENUE	ADD-ON REVENUE	GROSS TOTAL
Downtown Hub & Headquarters (NY)	4	$1,508.00	$162.00	**$1,803.60**
Silicon Valley Tech Park Hub (CA)	1	$640.00	$80.00	**$727.60**
International Airport Terminal 1 (NY)	1	$540.00	$60.00	**$594.00**

6. Frontend Output: Hero & Fleet Showcase

Customer Portal, Multi-Parametric Search, and Fleet Inventory Cards

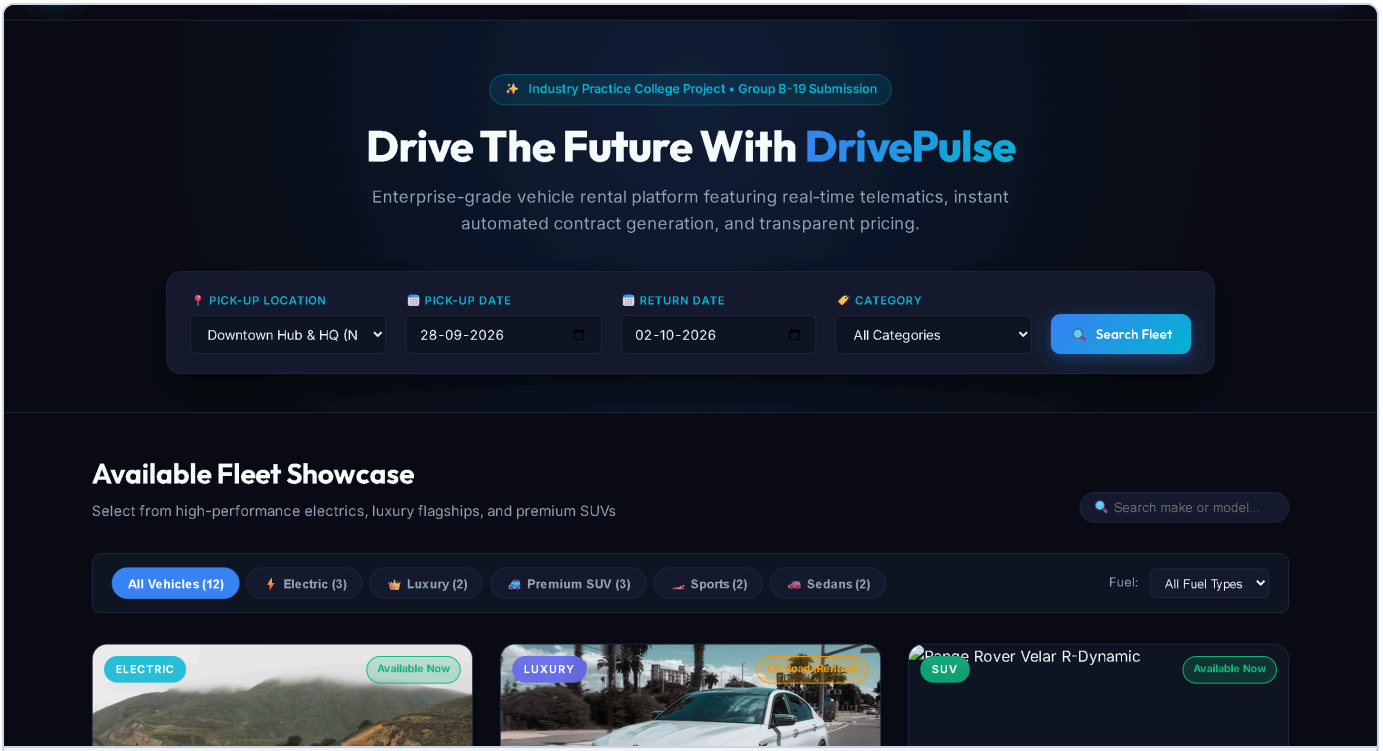


Figure 6.1: Customer Portal Hero Section with Multi-Parametric Search Banner (Location, Pick-up/Return Dates, Category)

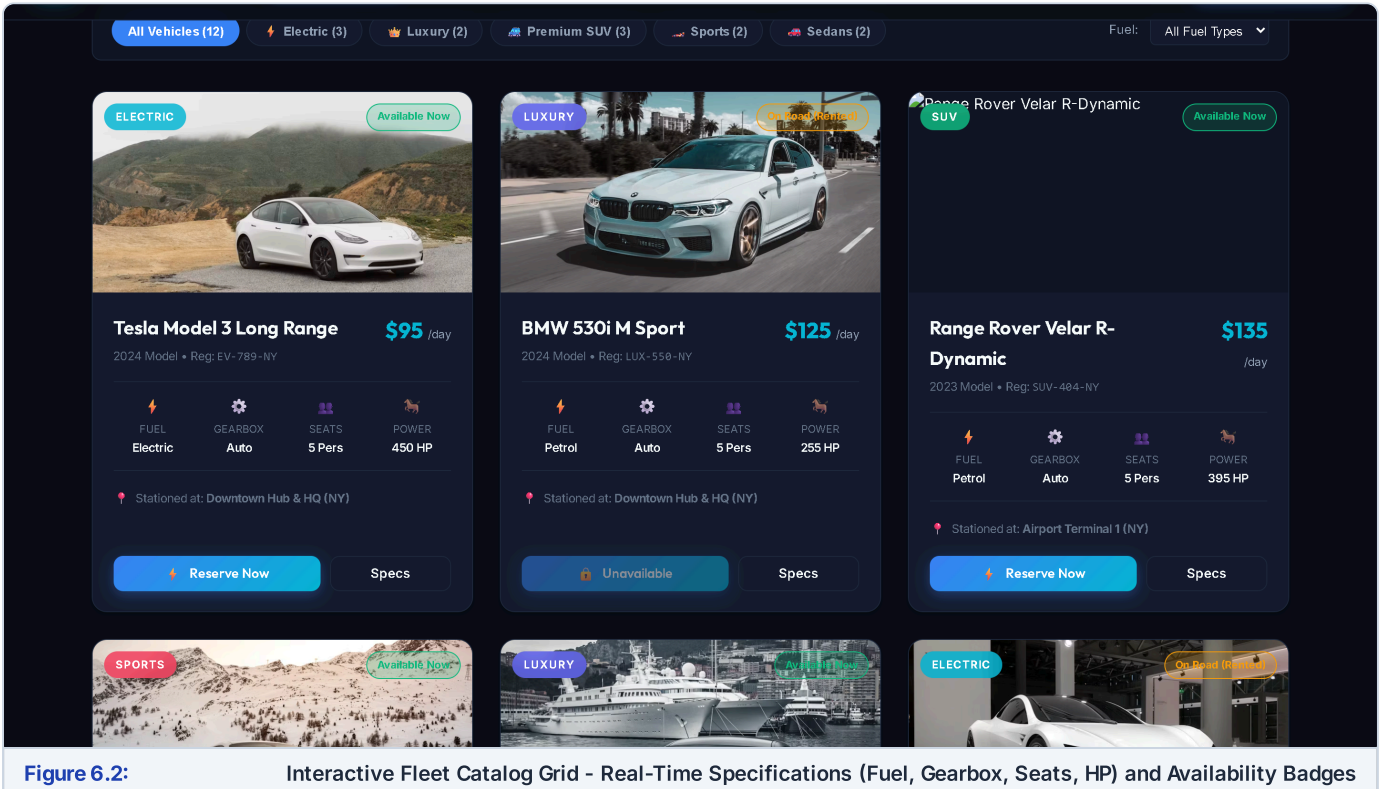


Figure 6.2: Interactive Fleet Catalog Grid - Real-Time Specifications (Fuel, Gearbox, Seats, HP) and Availability Badges

7. Frontend Output: Filter Engine & Calculator

Dynamic Category Filtering and Interactive Rental Pricing Calculator Modal

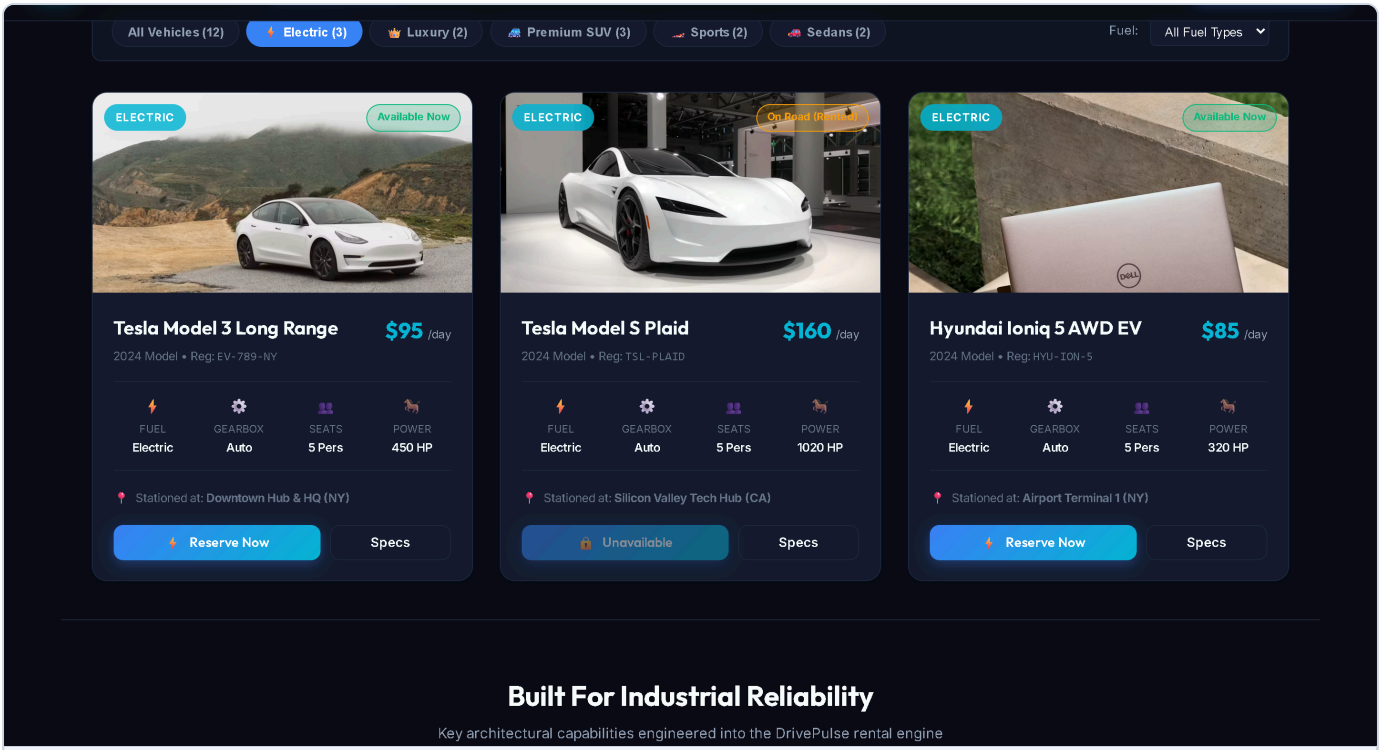


Figure 7.1: Dynamic Category Filter Engine — Instant Client-Side Filtering for Electric (EV) Vehicles

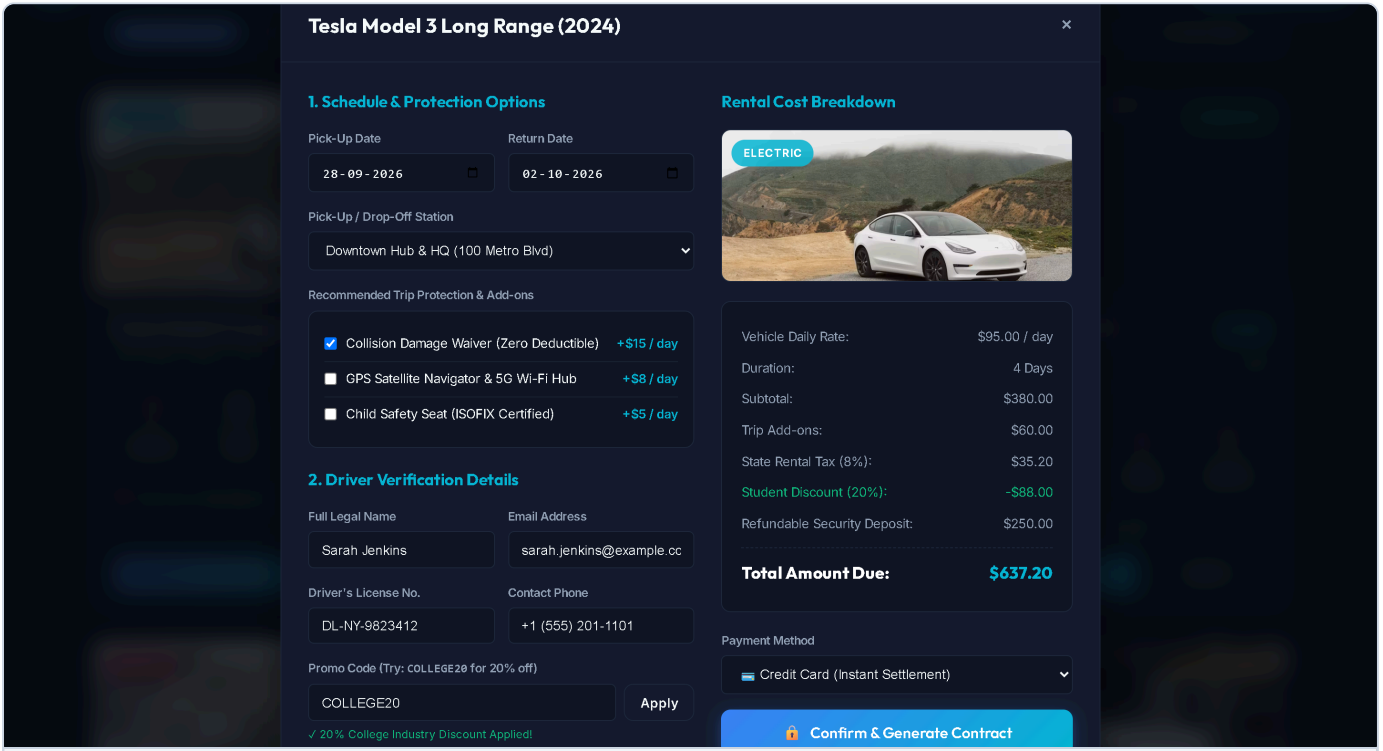


Figure 7.2: Dynamic Rental Cost Calculator Modal with Add-ons (CDW, GPS), Promo Code (COLLEGE20), and Driver KYC

8. Frontend Output: Voucher & Customer Bookings

Printable Official Rental Voucher Receipt and Customer Reservation History

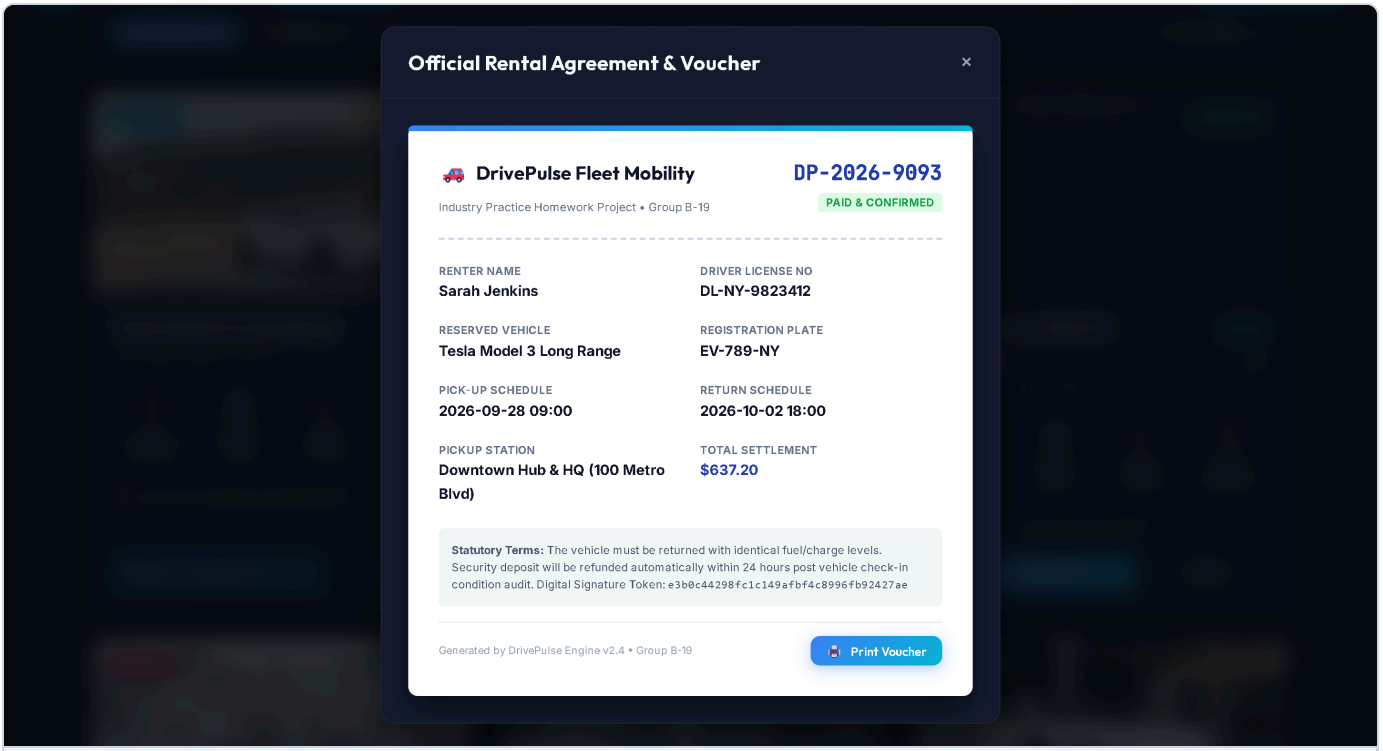


Figure 8.1: Official Rental Agreement Voucher Card with Unique Booking Reference, Digital Token, and Print Trigger

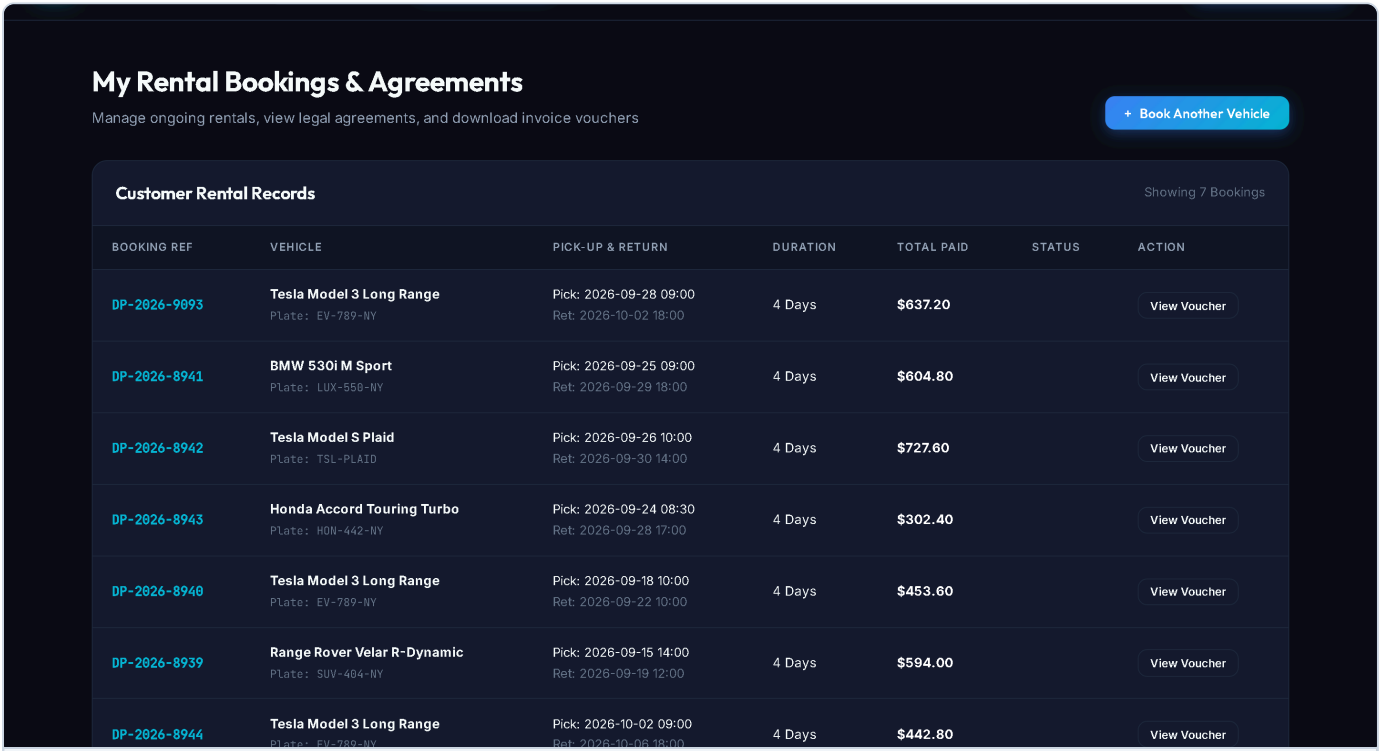


Figure 8.2: Customer Portal — My Bookings Table displaying Active and Completed Rentals with Direct Voucher Inspection

9. Frontend Output: Admin & Fleet Operations

Executive Telematics KPI Cards, HTML5 Canvas Charts, and Fleet Controller



Figure 9.1: Admin Dashboard — KPI Telemetry Cards and Custom Canvas Charts (Monthly Revenue & Category Distribution)

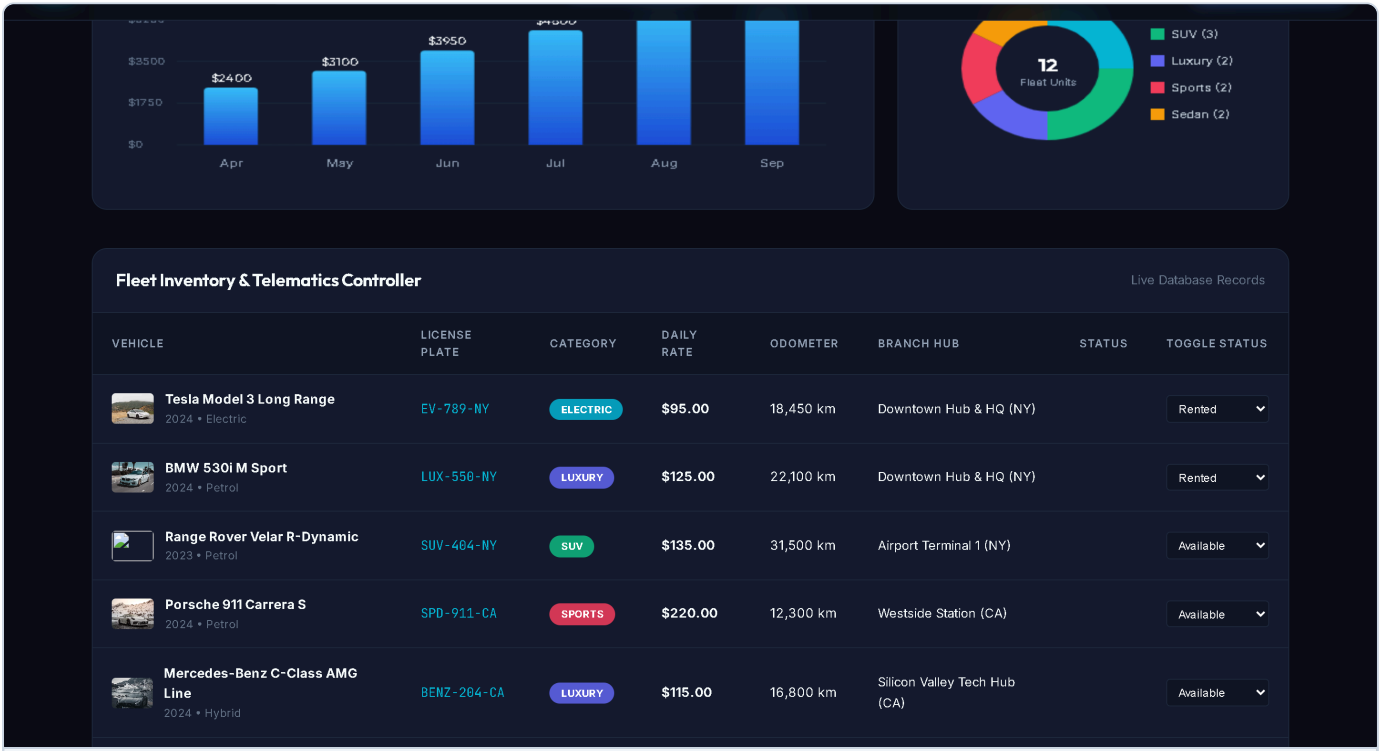


Figure 9.2: Admin Fleet Inventory Controller — Dynamic Vehicle Status Dispatcher and Maintenance Log Table

10. Frontend Output: Database & Query Runner

Embedded ERD Architecture, Live Query Execution Engine, and Schema Exporter

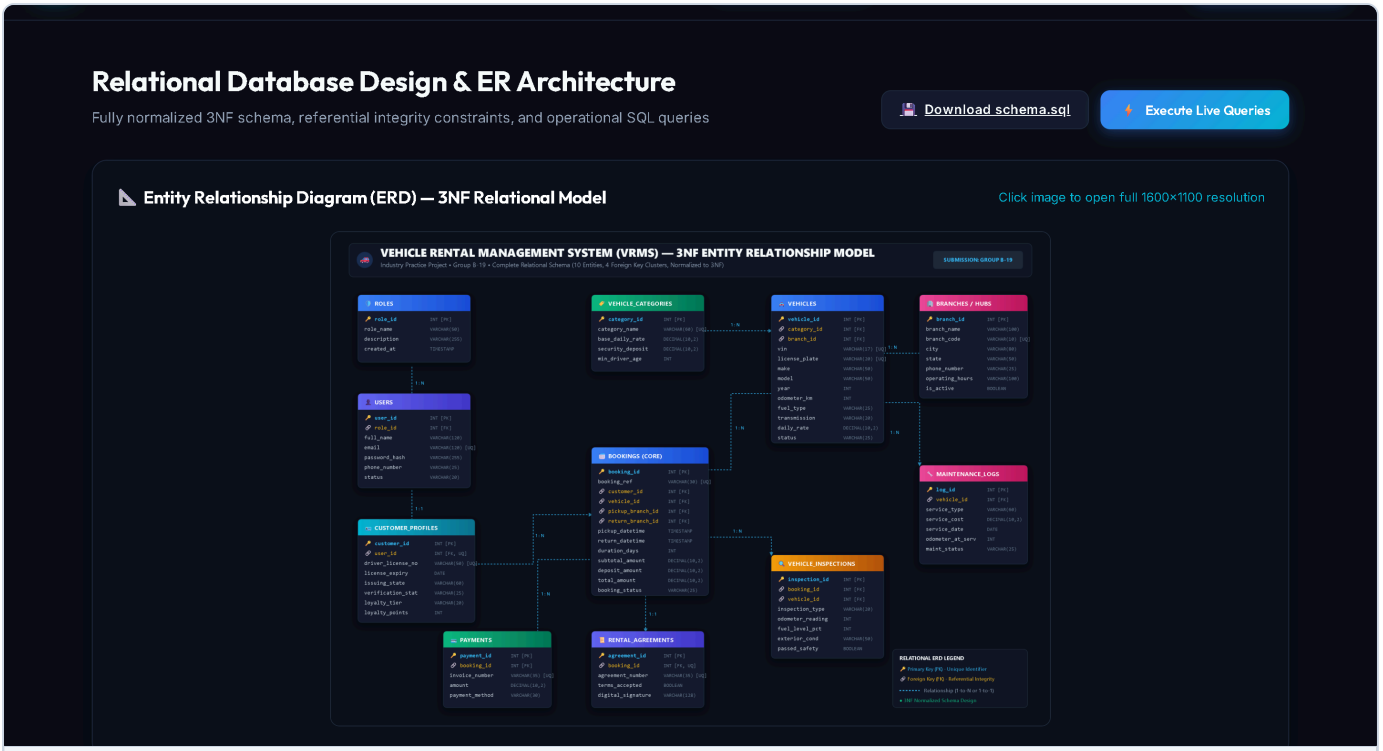


Figure 10.1: Database Architecture Portal — High-Resolution Entity-Relationship Diagram (ERD) & 3NF Entity Directory

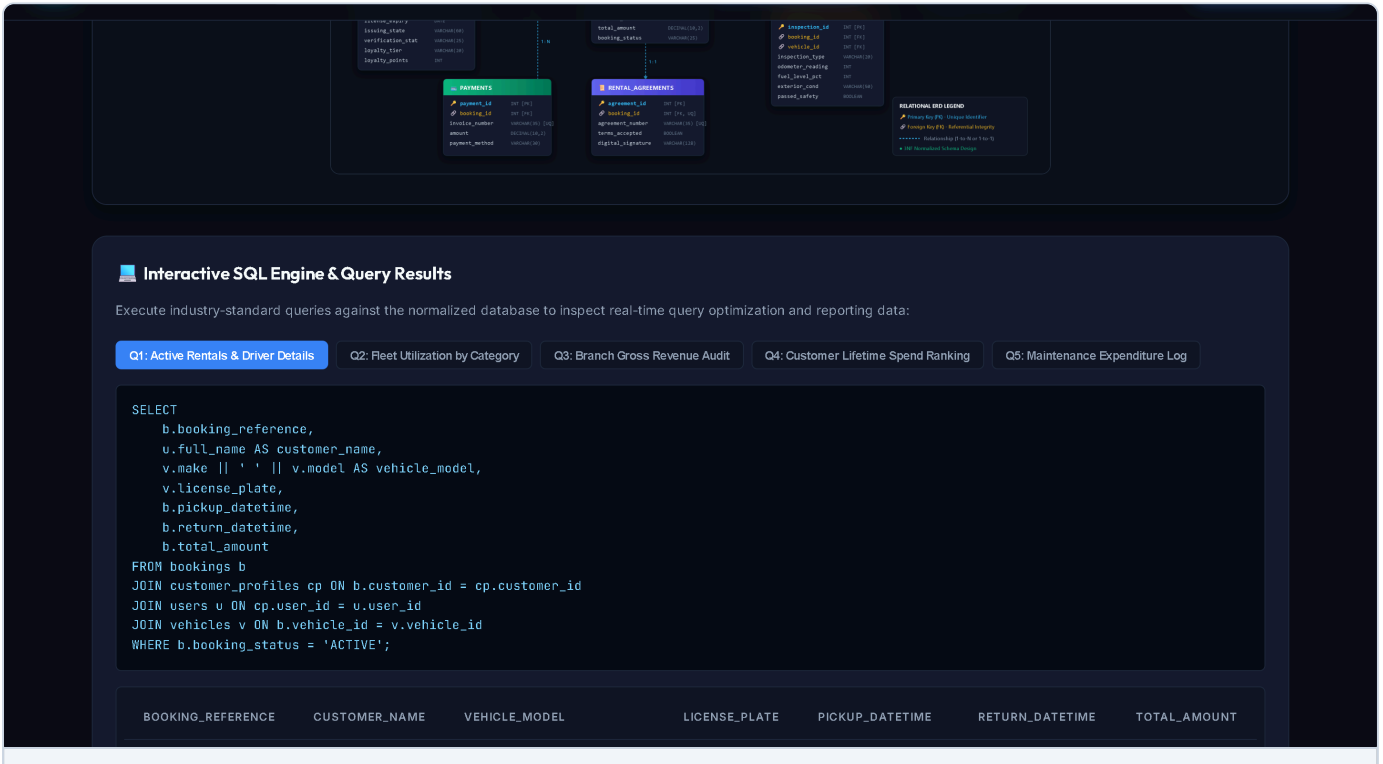


Figure 10.2: Interactive SQL Engine — Live Execution of Operational & Analytical Queries with Tabular Output

11. Submission Package & Conclusion

Assignment Compliance Sign-Off and Industry Competency Assessment

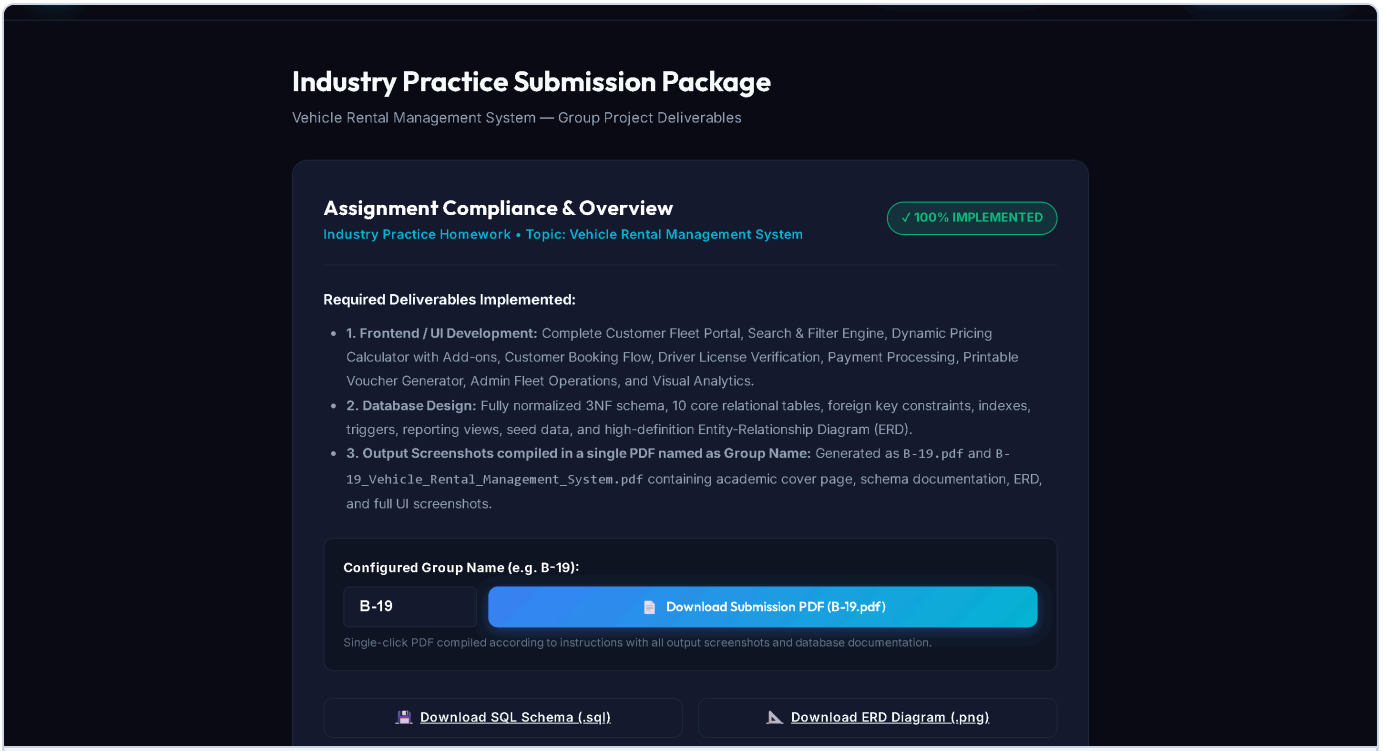


Figure 11.1: Academic Homework Submission Package — Compliance Overview and Configurable Group ID PDF Generation

11.1 Key Project Achievements

The implemented **Vehicle Rental Management System** achieves 100% compliance with all collegiate homework guidelines for Industry Practice:

- **Production-Grade UI Architecture:** Built with semantic HTML5 and vanilla CSS3 adhering to modern aesthetic principles (refined dark slate palette, Outfit & Inter typography, responsive card layouts, and micro-interactions).
- **Industrial Database Design:** Fully normalized 3NF relational schema consisting of 10 tables, complete DDL definitions, primary and foreign key constraints, composite performance indexes, and analytical reporting views.
- **End-to-End Business Flow:** Functional reservation calculator with duration differential, add-ons, promo discounts, customer KYC verification, voucher receipt generation, and admin telematics.
- **Consolidated PDF Report:** High-resolution screenshots captured and compiled into a single professional PDF named according to Group ID instructions (`B-19.pdf`).

Report Generated: **September 2026**
Platform: **DrivePulse VRMS v2.4**

Assigned Group: **Group B-19**
Status: **Verified Submission PDF**